

SAI VENKATA ADITHYA CHALLA

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<https://adithyachalla12.github.io/Resume/>

EDUCATION

University of Southern California

Aug 2024 - May 2026

Master's, Computer Science (GPA: 3.75/4.0)

- **Coursework:** Applied NLP, Database Systems, Information Retrieval, Machine Learning for Data Science, Web Technologies, Deep Learning, Analysis of Algorithms, Software Engineering

GITAM University

Jun 2019 - Apr 2023

Bachelor of Technology, Computer Science (GPA: 9.15/10)

- **Coursework:** Foundations of AI, Data Warehouse & Mining, Object Oriented Programming, Operating Systems, Agile Software Development

TECHNICAL SKILLS

- **Languages & Frameworks:** C, Java, Python, R, PySpark, KQL, MySQL, HTML, CSS, JavaScript, Flask, React, Node.js
- **Cloud & Data Engineering:** AWS S3, AWS EC2, AWS Lambda, AWS Quicksight, Azure Data Factory, Azure Synapse Analytics, Microsoft Fabric
- **Data Analysis & Visualization:** Numpy, Pandas, Matplotlib, Seaborn, Power BI, Power Automate, Tableau, D3.js
- **ML/AI Frameworks:** TensorFlow, PyTorch, Keras, Scikit-learn, NLTK, OpenCV, Streamlit, HuggingFace, Agent Development Kit (ADK), FastAPI, Transformers
- **DevOps & Tools:** Git, GitHub, Docker, REST APIs, CI/CD

EXPERIENCE

Vosyn | Machine Learning Engineer Intern

Feb 2026 - May 2026

- Developed and deployed machine learning models using Python and PyTorch, building scalable pipelines for data preprocessing, training, and inference
- Evaluated transcription quality across diverse audio datasets by analyzing word error rate (WER), language robustness, and noise handling to improve model reliability and deployment readiness
- Conducted benchmarking of Whisper and Canary models on diverse audio datasets, assessing transcription quality, latency, and language adaptability to support selection of optimal models for speech AI workflows

MAQ Software | Software Engineer I

Aug 2023 - Jun 2024

- Designed and maintained analytics datasets using Azure Data Factory, Databricks, and ADLS Gen2, enabling scalable reporting for SLA compliance, ticket aging, escalation trends, and operational performance monitoring
- Automated reporting workflows using Power Automate and Azure services, eliminating approximately 3 hours of daily manual effort and improving the timeliness and accuracy of operational reporting
- Developed self-service analytics solutions by integrating incident, telemetry, and access-management datasets using KQL and Azure Data Explorer, reducing engineering support requests by 40% and improving analyst productivity
- Conducted root-cause and trend analysis on operational data to identify recurring escalation drivers and service bottlenecks, providing actionable insights that improved issue resolution efficiency and service reliability

iAmX | AI/ML Intern

Feb 2022 - May 2023

- Developed machine learning models using Python and Scikit-learn to analyze customer purchasing behavior, forecast product demand, and identify sales trends from retail datasets
- Built product recommendation and item-matching solutions using TF-IDF, cosine similarity, and collaborative filtering techniques to improve product discovery and cross-selling opportunities
- Performed feature engineering, model training, hyperparameter tuning, and evaluation to improve predictive performance across customer segmentation and recommendation use cases
- Researched and implemented AI-driven approaches for personalized retail experiences, leveraging customer, product, and behavioral data to generate actionable business insights

PROJECTS

AI-Powered Roommate Discovery Platform

- Developed RoomSync, an AI-powered roommate discovery platform, using a scalable client-server architecture to support intelligent roommate matching and housing workflows
- Built and optimized user and lister profile systems that captured lifestyle preferences, housing expectations, and behavioral data for AI-driven compatibility recommendations
- Implemented real-time chat and connection features that enabled users to communicate for roommate coordination, housing discussions, and subleasing opportunities, improving user engagement and platform interaction

Multi-Agent AI Career Guidance System

- Implemented a multi-agent system architecture using Google ADK and Gemini LLMs, supporting parallel and sequential agent orchestration
- Developed tool-augmented LLM agents integrating external search APIs and currency conversion utilities for structured data retrieval and normalization
- Designed a modular data processing pipeline with parallel execution, aggregation layers, and standardized output schemas for downstream consumption

Knowledge Conflict Resolution System

- Built an interpretable multimodal knowledge conflict resolution system using CLIP, BLIP, and semantic similarity metrics to detect contradictions between text and images
- Trained a CNN-BiLSTM text backbone on 100K Fakeddit samples, achieving 0.83 accuracy with stable generalization and class-balanced performance
- Conducted evidence-grounded claim verification on the LIAR benchmark using BERT/roBERTa cross-encoders, reducing adjacent label confusion and improving calibration 0.18-0.07 to inform multimodal decision logic

Code Generation from API Documentation

- Analyzed instruction-to-API code generation on the API Pack dataset, identifying that zero-shot LLMs achieve near-zero exact-match accuracy despite producing syntactically valid code
- Improved API code generation reliability by fine-tuning StarCoder, Qwen, and GPT-2 using LoRA, achieving 100% syntax correctness and ~19 SacreBLEU, significantly outperforming zero-shot baselines